Applied Mathematics for Chemists

(화학수학)

Lecture 9

Differential Equations

12. Differential Equations

(12장 미분방정식)

Differential Equations (DE, 미분방정식)

Equations that contain derivatives

- Ordinary differential equation (ODE, 상미분방정식) is a differential equation of a single-variable function

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + \frac{\mathrm{d}y}{\mathrm{d}x} + x = 5$$

- Partial differential equation (PDE, 편미분방정식) contains partial derivatives

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} = 0$$

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Examples from Physical Chemistry

- Quantum mechanics: 1D Schrödinger equation

$$i\hbar\frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\Psi}{\partial x^2} + V\Psi$$

- Kinetics: second-order reaction

$$\frac{\mathrm{d}[\mathrm{A}]}{\mathrm{d}t} = -k[\mathrm{A}]^2$$

- Transport phenomena: 1D heat equation

$$\frac{\partial u}{\partial t} = k\frac{\partial^2 u}{\partial x^2}$$

Terminology

- Order of DE (계수): order of the highest derivative

- Linear (선형) vs. nonlinear (비선형)
  - Linear: contains the function (y) and its derivatives only to the first power (Note: terms in x can be nonlinear)

$$a_n(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n}+a_{n-1}(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}}+\cdots+a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x}+a_0(x)y=g(x)$$

  - Nonlinear: otherwise

$$(1-y)\frac{\mathrm{d}y}{\mathrm{d}x}+2y=e^x, \qquad \frac{\mathrm{d}^2 y}{\mathrm{d}x^2}+\sin y=0, \qquad \cdots$$

Linear Differential Equations

$$a_n(x)\frac{\mathrm{d}^n y}{\mathrm{d}x^n} + a_{n-1}(x)\frac{\mathrm{d}^{n-1} y}{\mathrm{d}x^{n-1}} + \cdots + a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y = g(x)$$

- Homogeneous (동차): $g(x) = 0$
- Inhomogeneous (비동차): $g(x) \neq 0$

General and Particular Solutions
(일반해와 특수해)

- Solution: any function that reduces the DE to an identity when substituted into the DE

For a DE of order n,

- General solution: contains n arbitrary constants
- Particular solution: contains no arbitrary constants
  - Particular solutions can be generated from the general solution

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Example

$$\frac{d^4y}{dx^4} + 2\frac{d^2y}{dx^2} + y = 0$$

- $y = \cos x$ is a particular solution
- $y = -x \sin x$ is another particular solution

- $y = c_1 \cos x + c_2 \sin x + c_3 x \cos x + c_4 x \sin x$ is a general solution

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Initial-Value Problems (IVP, 초깃값 문제)

- For a DE of order n, if n additional conditions are given, one can determine the arbitrary constants of the general solution
- Initial conditions (IC, 초기 조건): values of the function and its first $(n-1)$ derivatives at a single point

$$y(x_0) = y_0, \qquad \left.\frac{\mathrm{d}y}{\mathrm{d}x}\right|_{x=x_0} = y_1, \qquad \cdots , \qquad \left.\frac{\mathrm{d}^{n-1}y}{\mathrm{d}x^{n-1}}\right|_{x=x_0} = y_{n-1}$$

Example

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 4y = 0$$

General solution:

$$y = c_1 \cos 2x + c_2 \sin 2x$$

Initial conditions:

$$y(0) = 1, \qquad \mathrm{d}y/\mathrm{d}x\big|_{x=0} = -6$$

Solution of the IVP:

$$y = \cos 2x - 3 \sin 2x$$

Boundary-Value Problems (BVP, 경곗값 문제)

- For a DE of order n, if n additional conditions are given, one can determine the arbitrary constants of the general solution

- Boundary conditions (BC, 경계 조건): values of the function and its derivatives at different points

Example

$$\frac{\mathrm{d}^2 y}{\mathrm{d}x^2} + 4y = 0$$

General solution:

$$y = c_1 \cos 2x + c_2 \sin 2x$$

Boundary conditions:

$$y(0) = 1, \qquad y(\pi/4) = -2$$

Solution of the BVP:

$$y = \cos 2x - 2 \sin 2x$$

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Solving a Differential Equation

1. Obtain the general solution: hardest part!
2. If initial/boundary conditions are given, substitute the conditions to determine the coefficients

- No universal method to solve all DEs
- For specific forms of DEs, we have general solving methods

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First-Order ODEs

$$F\left(y, \frac{\mathrm{d}y}{\mathrm{d}x}\right) = 0$$

- Separable variables (§ 12.4)
- Exact equations (§ 12.5, 12.6)
- Linear equations (not in Mortimer)

Separable Variables

- Separable DE:

$$g(y)\frac{\mathrm{d}y}{\mathrm{d}x} = f(x)$$

This DE can be manipulated into

$$g(y)\,\mathrm{d}y = f(x)\,\mathrm{d}x$$

$$\int g(y)\,\mathrm{d}y = \int f(x)\,\mathrm{d}x + C$$

This equation provides the general solution

Example 12.9

In a first-order chemical reaction with no back reaction, the concentration of the reactant is governed by

$$-\frac{\mathrm{d}c}{\mathrm{d}t} = kc$$

where c is the concentration, t is the time, and k is the rate constant. Solve the equation to find c as a function of t.

Rearrange the equation to separate variables:

$$\frac{1}{c}\,\mathrm{d}c = -k\;\mathrm{d}t$$

$$\int \frac{1}{c}\,\mathrm{d}c = -\int k\;\mathrm{d}t$$

Example 12.9

$$-\frac{\mathrm{d}c}{\mathrm{d}t} = kc$$

Solve the equation to find c as a function of t.

$$\int \frac{1}{c}\,\mathrm{d}c = -\int k\,\mathrm{d}t$$

Integration leads to

$$\ln c = -kt + C$$

$$c = e^{-kt+C} = Ae^{-kt}$$

This is the general solution and A is an arbitrary constant.

Exact Differential Equations

- Pfaffian form:

$$M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0$$

If the differential equation is exact,

there is a function $f(x, y)$ such that

$$\mathrm{d}f = M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0$$

which implies that $f(x, y) = k$ (constant)

How to Find $\boldsymbol{f(x,y)}$

$$\mathrm{d}f = M(x,y)\,\mathrm{d}x + N(x,y)\,\mathrm{d}y = 0$$

$$f(x_1,y_1) = f(x_0,y_0) + \int_C \mathrm{d}f$$

where C is a curve connecting (x_0,y_0) and (x_1,y_1). Take a rectangular path from (x_0,y_0) to (x_1,y_0) to (x_1,y_1):

$$f(x_1,y_1) = f(x_0,y_0) + \int_{x_0}^{x_1} M(x,y_0)\,\mathrm{d}x + \int_{y_0}^{y_1} N(x_1,y)\,\mathrm{d}y$$

This gives the relationship between x and y.

Example 12.10

Solve the differential equation

$$2xy\,\mathrm{d}x + x^2\mathrm{d}y = 0.$$

Check the exactness first:

$$\frac{\partial(2xy)}{\partial y} = 2x, \qquad \frac{\partial(x^2)}{\partial x} = 2x$$

Do a line integral from (x_0, y_0) to (x_1, y_0) to (x_1, y_1):

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0)\,\mathrm{d}x + \int_{y_0}^{y_1} N(x_1, y)\,\mathrm{d}y$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} (2xy_0)\,\mathrm{d}x + \int_{y_0}^{y_1} (x_1^2)\,\mathrm{d}y$$

Example 12.10

Solve the differential equation

$$2xy\,\mathrm{d}x + x^2\mathrm{d}y = 0.$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} (2xy_0)\,\mathrm{d}x + \int_{y_0}^{y_1} (x_1^2)\,\mathrm{d}y$$

$$f(x_1, y_1) = f(x_0, y_0) + (x_1^2 y_0 - x_0^2 y_0) + (x_1^2 y_1 - x_1^2 y_0)$$

Hence,

$$f(x_1, y_1) - f(x_0, y_0) = x_1^2 y_1 - x_0^2 y_0$$

which implies that $f(x, y) = x^2 y$.

Example 12.10

Solve the differential equation

$$2xy\,\mathrm{d}x + x^2\mathrm{d}y = 0.$$

$$f(x, y) = x^2y$$

Now, we impose $f(x, y) = k$:

$$x^2y = k$$

Therefore,

$$y = \frac{k}{x^2}$$

This is the general solution and k is an arbitrary constant.

Integrating Factors

- Pfaffian form:

$$M(x, y)\,\mathrm{d}x + N(x, y)\,\mathrm{d}y = 0$$

If the differential equation is inexact, but we know the integration factor $g(x, y)$ that makes

$$g(x, y)M(x, y)\,\mathrm{d}x + g(x, y)N(x, y)\,\mathrm{d}y = 0$$

an exact differential equation, we can solve this DE.

Example 12.11

Solve the differential equation

$$y\,\mathrm{d}x - x\,\mathrm{d}y = 0$$

where $1/x^2$ is an integrating factor.

Check the exactness first:

$$\frac{\partial(y)}{\partial y} = 1, \qquad \frac{\partial(-x)}{\partial x} = -1$$

Hence, the differential equation is inexact.

Multiplying the integrating factor, we obtain

$$\frac{y}{x^2}\,\mathrm{d}x - \frac{1}{x}\,\mathrm{d}y = 0$$

Example 12.11

Solve the differential equation

$$y \, \mathrm{d}x - x \, \mathrm{d}y = 0$$

where $1/x^2$ is an integrating factor.

$$\frac{y}{x^2} \, \mathrm{d}x - \frac{1}{x} \, \mathrm{d}y = 0$$

Check the exactness:

$$\frac{\partial}{\partial y}\left(\frac{y}{x^2}\right) = \frac{1}{x^2}, \qquad \frac{\partial}{\partial x}\left(-\frac{1}{x}\right) = \frac{1}{x^2}$$

Now the differential equation is exact.

Example 12.11

Solve the differential equation

$$y\,\mathrm{d}x - x\,\mathrm{d}y = 0$$

where $1/x^2$ is an integrating factor.

$$\frac{y}{x^2}\,\mathrm{d}x - \frac{1}{x}\,\mathrm{d}y = 0$$

Do a line integral from (x_0, y_0) to (x_1, y_0) to (x_1, y_1):

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} M(x, y_0)\,\mathrm{d}x + \int_{y_0}^{y_1} N(x_1, y)\,\mathrm{d}y$$

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} \left(\frac{y_0}{x^2}\right)\mathrm{d}x + \int_{y_0}^{y_1} \left(-\frac{1}{x_1}\right)\mathrm{d}y$$

Example 12.11

Solve the differential equation

$$y\,\mathrm{d}x - x\,\mathrm{d}y = 0$$

where $1/x^2$ is an integrating factor.

$$f(x_1, y_1) = f(x_0, y_0) + \int_{x_0}^{x_1} \left(\frac{y_0}{x^2}\right) \mathrm{d}x + \int_{y_0}^{y_1} \left(-\frac{1}{x_1}\right) \mathrm{d}y$$

$$f(x_1, y_1) = f(x_0, y_0) + y_0\left(-\frac{1}{x_1} + \frac{1}{x_0}\right) - \frac{1}{x_1}(y_1 - y_0)$$

$$f(x_1, y_1) - f(x_0, y_0) = \frac{y_0}{x_0} - \frac{y_1}{x_1}$$

This implies that $f(x, y) = -y/x$.

Example 12.11

Solve the differential equation

$$y\,\mathrm{d}x - x\,\mathrm{d}y = 0$$

where $1/x^2$ is an integrating factor.

$$f(x, y) = -y/x$$

Now, we impose $f(x, y) = k$:

$$-y/x = k$$

Therefore,

$$y = k'x$$

This is the general solution and k' is an arbitrary constant.

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Linear Equations

$$a_1(x)\frac{\mathrm{d}y}{\mathrm{d}x} + a_0(x)y = g(x)$$

- Standard form:

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = f(x)$$

where $P(x) = a_0(x)/a_1(x)$ and $f(x) = g(x)/a_1(x)$

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Solving the Standard Form

$$\frac{\mathrm{d}y}{\mathrm{d}x} + P(x)y = f(x)$$

Multiply both sides by $e^{\int P(x)\mathrm{d}x}$;

$$e^{\int P(x)\mathrm{d}x}\frac{\mathrm{d}y}{\mathrm{d}x} + e^{\int P(x)\mathrm{d}x}P(x)y = e^{\int P(x)\mathrm{d}x}f(x)$$

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[e^{\int P(x)\mathrm{d}x}y\right] = e^{\int P(x)\mathrm{d}x}f(x)$$

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Solving the Standard Form

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[e^{\int P(x)\mathrm{d}x}y\right] = e^{\int P(x)\mathrm{d}x}f(x)$$

Integration leads to

$$e^{\int P(x)\mathrm{d}x}y = \int e^{\int P(x)\mathrm{d}x}f(x)\,\mathrm{d}x + C$$

Hence,

$$y = e^{-\int P(x)\mathrm{d}x}\int e^{\int P(x)\mathrm{d}x}f(x)\,\mathrm{d}x + Ce^{-\int P(x)\mathrm{d}x}$$

Example

Solve the differential equation

$$3\frac{\mathrm{d}y}{\mathrm{d}x} + 12y = 4.$$

The standard form is

$$\frac{\mathrm{d}y}{\mathrm{d}x} + 4y = \frac{4}{3}$$

and $P(x) = 4$, which gives $e^{\int P(x)\mathrm{d}x} = e^{\int 4\mathrm{d}x} = e^{4x}$.

$$e^{4x}\frac{\mathrm{d}y}{\mathrm{d}x} + 4e^{4x}y = \frac{4}{3}e^{4x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^{4x}y) = \frac{4}{3}e^{4x}$$

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Example

Solve the differential equation

$$3\frac{\mathrm{d}y}{\mathrm{d}x} + 12y = 4.$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^{4x}y) = \frac{4}{3}e^{4x}$$

Integration leads to

$$e^{4x}y = \frac{4}{3}\int e^{4x}\,\mathrm{d}x + C = \frac{1}{3}e^{4x} + C$$

Hence,

$$y = \frac{1}{3} + Ce^{-4x}$$

Another Example

Solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y = x$$

subject to $y(0) = 4$.

The equation is in standard form and

$P(x) = 1$, which gives $e^{\int P(x)\mathrm{d}x} = e^{\int 1\mathrm{d}x} = e^{x}$.

$$e^{x}\frac{\mathrm{d}y}{\mathrm{d}x} + e^{x}y = xe^{x}$$

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^{x}y) = xe^{x}$$

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Another Example

Solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y = x$$

subject to $y(0) = 4$.

$$\frac{\mathrm{d}}{\mathrm{d}x}(e^x y) = xe^x$$

Integration leads to

$$e^x y = \int xe^x \,\mathrm{d}x + C = xe^x - e^x + C$$

Hence,

$$y = x - 1 + Ce^{-x}$$

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Another Example

Solve the differential equation

$$\frac{\mathrm{d}y}{\mathrm{d}x} + y = x$$

subject to $y(0) = 4$.

$$y = x - 1 + Ce^{-x}$$

The initial condition states that for $x = 0$, $y = 4$. Substitution gives

$$4 = 0 - 1 + C$$

Thus, $C = 5$ and the solution is

$$y = x - 1 + 5e^{-x}$$